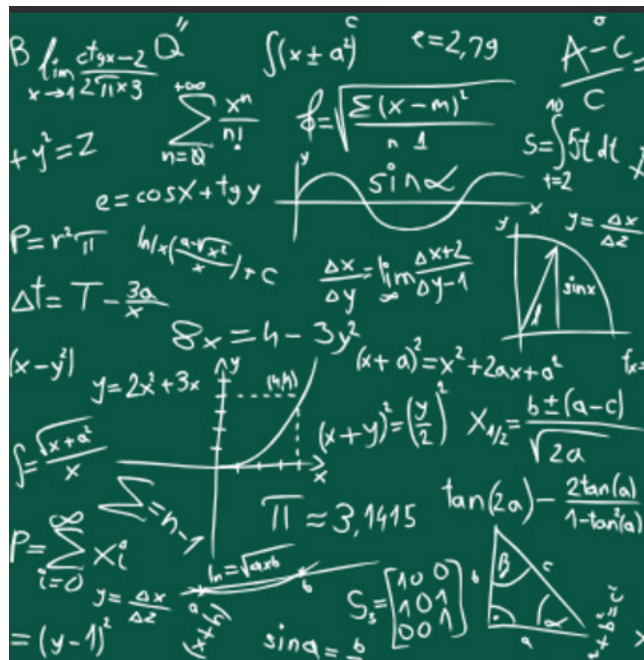


Number sets and intervals



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 The five main number sets

1.1 The natural numbers

1.2 The integers

$\mathbb{Z}$ is the set of **integers** (also called signed numbers), that is:

$$0, 1, -1, 2, -2, \dots$$

1.3 The rational numbers

$\mathbb{Q}$ is the set of **rational numbers**, that is, of the fractions:

$$\frac{n}{m}, \text{ where } n, m \in \mathbb{Z}, \text{ and } m \neq 0.$$

Rational numbers can also be written in decimal form. For example:

$$\frac{3}{4} = 0,75, \text{ while } \frac{1}{3} = 0,333\dots = 0,\bar{3}.$$

A rational number, written in decimal form, may have after the decimal point a finite number of digits (different from zero), or an infinite number of digits different from zero, which however repeat *periodically*.

Remark 1

$$0,\bar{9} = 1$$

There are different proofs of this remark, based on different mathematical techniques.

Proof. A simple proof follows directly from the definition of 1 divided by 3; indeed, we have:

$$\begin{aligned}\frac{1}{3} &= 0,\bar{3} \\ \frac{1}{3} \cdot 3 &= 0,\bar{3} \cdot 3 \\ 1 &= 0,\bar{9}\end{aligned}$$

□

Proof. Using algebraic arguments we can write:

$$\begin{array}{ll}
 x = 0.999\dots & \\
 10x = 9.999\dots & \text{multiplying by 10} \\
 10x = 9 + 0.999\dots & \text{separating the integer part from the fractional part} \\
 10x = 9 + x & \text{by definition of } x \\
 9x = 9 & \text{subtracting } x \\
 x = 1 & \text{dividing by 9}
 \end{array}$$

□

Proof. A proof by contradiction is the following:

$$\begin{array}{l}
 0.\bar{9} \neq 1 \\
 0.\bar{9} \cdot 9 \neq 1 \cdot 9 \\
 0.\bar{9} \cdot 9 + 0.\bar{9} \neq 1 \cdot 9 + 0.\bar{9} \\
 0.\bar{9} \cdot 9 + 0.\bar{9} \neq 9.\bar{9} \\
 0.\bar{9} \cdot (9 + 1) \neq 9.\bar{9} \\
 0.\bar{9} \cdot (10) \neq 9.\bar{9} \\
 9.\bar{9} \neq 9.\bar{9} \quad \text{contradiction!}
 \end{array}$$

□

Other proofs start from the assumption that two numbers are identical if and only if their difference is equal to zero, and they are based on computing the value of $1 - 0.\bar{9}$.

Proof. We write the number $0.999\dots$ with n digits after the decimal point as $0.(9)_n$, hence $0.(9)_1 = 0.9$, $0.(9)_2 = 0.99$, $0.(9)_3 = 0.999$, and so on.

Given $\frac{1}{10^n} = 0.0\dots 01$, with n digits after the decimal point, the addition rules for decimal numbers imply

$$0.(9)_n + \frac{1}{10^n} = 1 \text{ moreover } 0.(9)_n < 1, \forall n \in \mathbb{N}.$$

We must prove that 1 is the smallest number that is not less than all the $0.(9)_n$. For this it is enough to prove that, if a number x is not greater than 1 and not less than all the $0.(9)_n$, then $x = 1$.

So let x be such that

$$0.(9)_n \leq x \leq 1$$

for every positive integer n . Hence

$$1 - 1 \leq 1 - x \leq 1 - 0.(9)_n$$

which, using basic arithmetic and the first equality established above, simplifies to

$$0 \leq 1 - x \leq \frac{1}{10^n}$$

This implies that the difference between 1 and x is less than the reciprocal of any positive integer. Hence this difference must be zero, and therefore $x = 1$; which in turn implies

$$0.999\dots = 1$$

□

This proof is based on the fact that 0 is the only non-negative number less than all the reciprocals of the positive integers, or equivalently that there is no number greater than every integer.

This is the **Archimedean property**, which holds for the rational and the real numbers.

1.4 The real numbers

$\mathbb{R}$ is the set of **real numbers**, that is, those which, written in decimal form, have after the decimal point any sequence of digits different from zero, possibly even *infinite* and *non-periodic*.

That numbers of this last kind exist (that is, real but not rational), can be understood by reflecting on examples such as:

$$0,10110111011110\dots$$

After the decimal point, the previous number has: one digit equal to 1, then 0, then two digits equal to 1, then 0, then three digits equal to 1 ... and so on. It is clear that this rule defines a decimal number precisely. On the other hand, the string of nonzero digits after the decimal point is neither finite nor periodic: therefore this number is irrational.

1.5 The complex numbers

$\mathbb{C}$ is the set of **complex numbers**, that is, numbers of the form $a + i b$, where a, b are real numbers, and i is the *imaginary unit*, that is, a number whose square is -1 .

Among the number sets introduced, the following inclusions hold:

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}.$$

As indicated by the symbols, all the inclusions are strict:

- there exist integers that are not natural numbers (the negative numbers),
- there exist rational numbers that are not integers (the proper fractions),
- there exist real numbers that are not rational (the irrational numbers),
- there exist complex numbers that are not real (the imaginary numbers).

2 Intervals

Definition 1: of interval

Given two real numbers a, b , an **interval** with endpoints a and b is one of the following sets:

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}, \quad [a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}, \quad (a, b) = \{x \in \mathbb{R} : a < x < b\}$$

- As can be seen, the square (round) bracket at one of the two endpoints indicates that this endpoint is included in (excluded from) the interval.

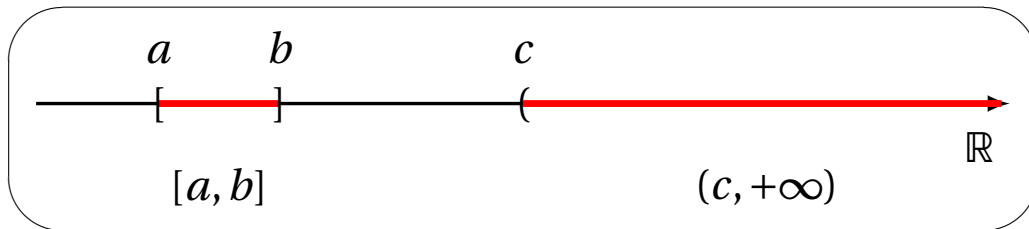
- The intervals $[a, b]$ are called **closed**; the intervals (a, b) are called **open**.
- All the intervals listed above are bounded; half-lines are also called (unbounded) intervals, for example:

$$(-\infty, b) = \{x \in \mathbb{R} : x < b\}$$

$$[a, +\infty) = \{x \in \mathbb{R} : x \geq a\}$$

or the whole line

$$\mathbb{R} = (-\infty, +\infty)$$



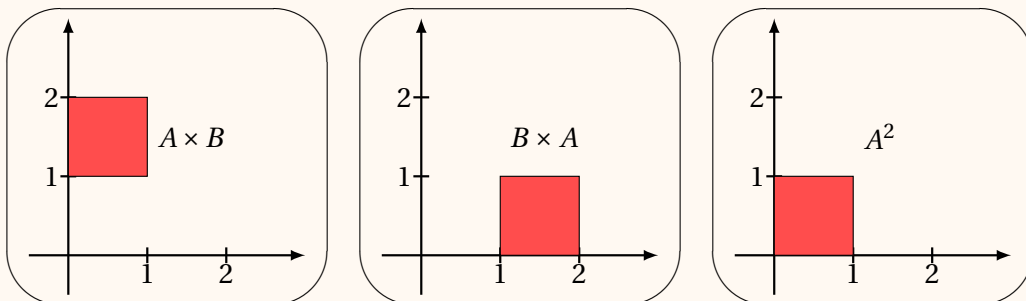
It can be proved that the intervals, bounded or unbounded, are exactly the subsets I of $\mathbb{R}$ that satisfy the following property (called **connectedness**):

$$x_1 < x_2 < x_3, \text{ if } x_1, x_3 \in I, \text{ then } x_2 \in I$$

- In what follows we will sometimes consider the Cartesian product of two (or more) intervals, which can be given the geometric meaning of a rectangle (in two dimensions) or of a parallelepiped (in three dimensions).

Example 1: Cartesian product of intervals

Let $A = [0, 1]$, $B = [1, 2]$; the figure



illustrates the sets $A \times B$, $B \times A$, $A \times A$, also denoted by A^2 . In general, $A \times B$ is different from $B \times A$.